Q1- What would be the IUPAC name and symbol for the element with atomic number 120?

Q2- How would you justify the presence of 18 elements in the 5th period of the Periodic Table?

Q3- The elements Z = 117 and 120 have not yet been discovered. In which family/group would you place these elements and also give the electronic configuration in each case.

Q4- Considering the atomic number and position in the periodic table, arrange the following elements in the increasing order of metallic character : Si, Be, Mg, Na, P.

Q5- Which of the following species will have the largest and the smallest size? Mg, Mg^{2+}, Al, Al^{3+}.

Q6- The first ionization enthalpy (Δ_i H) values of the third period elements, Na, Mg and Si are respectively 496, 737 and 786 kJ mol^{-1}. Predict whether the first Δ_i H value for Al will be more close to 575 or 760 kJ mol^{-1} ? Justify your answer.

Q7- Which of the following will have the most negative electron gain enthalpy and which the least negative? P, S, Cl, F. Explain your answer

Q8- Using the Periodic Table, predict the formulas of compounds which might be formed by the following pairs of elements; (a) silicon and bromine (b) aluminium and sulphur

Q9- Are the oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ same ?

Q10- Show by a chemical reaction with water that Na_2O is a basic oxide and Cl_2O_7 is an acidic oxide.

Q11- What is the basic theme of organisation in the periodic table?

Q12- Which important property did Mendeleev use to classify the elements in his periodic table and did he stick to that?

Q13- What is the basic difference in approach between the Mendeleev's Periodic Law and the Modern Periodic Law?

Q14- On the basis of quantum numbers, justify that the sixth period of the periodic table should have 32 elements.

Q15- In terms of period and group where would you locate the element with Z =114?

Q16- Write the atomic number of the element present in the third period and seventeenth group of the periodic table.

Q17- Which element do you think would have been named by (i) Lawrence Berkeley Laboratory (ii) Seaborg's group?

Q18- Why do elements in the same group have similar physical and chemical properties?

Q19- What does atomic radius and ionic radius really mean to you?

Q20- How do atomic radius vary in a period and in a group? How do you explain the variation?

Q21- What do you understand by isoelectronic species? Name a species that will be isoelectronic with each of the following atoms or ions. (i) F^- (ii) Ar (iii) Mg^{2+} (iv) Rb^+

Q22- Consider the following species : N^{3-}, O^{2-}, F^- , Na^+, Mg^{2+} and Al^{3+} (a) What is common in them? (b) Arrange them in the order of increasing ionic radii.

Q23- Explain why cation are smaller and anions larger in radii than their parent atoms?

Q24- What is the significance of the terms — 'isolated gaseous atom' and 'ground state' while defining the ionization enthalpy and electron gain enthalpy? Hint : Requirements for comparison purposes.

Q25- Energy of an electron in the ground state of the hydrogen atom is -2.18×10^{-18}J. Calculate the ionization enthalpy of atomic hydrogen in terms of J mol^{-1}. Hint: Apply the idea of mole concept to derive the answer.

Q26- Among the second period elements the actual ionization enthalpies are in the order Li < B < Be < C < O < N < F < Ne. Explain why (i) Be has higher Δ_i H than B (ii) O has lower Δ_i H than N and F?

Q27- How would you explain the fact that the first ionization enthalpy of sodium is lower than that of magnesium but its second ionization enthalpy is higher than that of magnesium?

Q28- What are the various factors due to which the ionization enthalpy of the main group elements tends to decrease down a group?

Q29- The first ionization enthalpy values (in kJ mol^{-1}) of group 13 elements are :

B	Al	Ga	In	Tl
801	577	579	558	589

How would you explain this deviation from the general trend ?

Q30- Which of the following pairs of elements would have a more negative electron gain enthalpy? (i) O or F (ii) F or Cl

Q31- Would you expect the second electron gain enthalpy of O as positive, more negative or less negative than the first? Justify your answer.

Q32- What is the basic difference between the terms electron gain enthalpy and electronegativity?

Q33- How would you react to the statement that the electronegativity of N on Pauling scale is 3.0 in all the nitrogen compounds?

Q34- Describe the theory associated with the radius of an atom as it (a) gains an electron (b) loses an electron Would you expect the first ionization enthalpies for two isotopes of the same element to be the same or different? Justify your answer.

Q35- What are the major differences between metals and non-metals?

Q36- Use the periodic table to answer the following questions. (a) Identify an element with five electrons in the outer subshell. (b) Identify an element that would tend to lose two electrons. (c) Identify an element that would tend to gain two electrons. (d) Identify the group having metal, non-metal, liquid as well as gas at the room temperature.

Q37- The increasing order of reactivity among group 1 elements is Li < Na < K < Rb Cl > Br > I. Explain.

Q38- Write the general outer electronic configuration of s-, p-, d- and f- block elements.

Q39- Assign the position of the element having outer electronic configuration (i) ns^2np^4 for n=3 (ii) $(n-1)d^2ns^2$ for n=4, and (iii) $(n-2)f^7(n-1)d^1ns^2$ for n=6, in the periodic table.

Q40- The first (Δ_iH_1) and the second (Δ_iH_2) ionization enthalpies (in kJ mol^{-1}) and the ($\Delta_{eg}H$) electron gain enthalpy (in kJ mol^{-1}) of a few elements are given below:

Elements	ΔH_1	ΔH_2	$\Delta_{eg}H$
I	520	7300	–60
II	419	3051	–48
III	1681	3374	–328
IV	1008	1846	–295
V	2372	5251	+48
VI	738	1451	–40

Q41- Which of the above elements is likely to be : (a) the least reactive element. (b) the most reactive metal. (c) the most reactive non-metal. (d) the least reactive non-metal. (e) the metal which can form a stable binary halide of the formula MX_2(X=halogen). (f) the metal which can form a predominantly stable covalent halide of the formula MX (X=halogen)?

Q42- Predict the formulas of the stable binary compounds that would be formed by the combination of the following pairs of elements. (a) Lithium and oxygen (b) Magnesium and nitrogen (c) Aluminium and iodine (d) Silicon and oxygen (e) Phosphorus and fluorine (f) Element 71 and fluorine

Q43- In the modern periodic table, the period indicates the value of : (a) atomic number (b) atomic mass (c) principal quantum number (d) azimuthal quantum number.

Q44- Which of the following statements related to the modern periodic table is incorrect?
(a) The p-block has 6 columns, because a maximum of 6 electrons can occupy all the orbitals in a p-shell. (b) The d-block has 8 columns, because a maximum of 8 electrons can occupy all the orbitals in a d-subshell.

(c) Each block contains a number of columns equal to the number of electrons that can occupy that subshell.

(d) The block indicates value of azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration.

Q45- Anything that influences the valence electrons will affect the chemistry of the element. Which one of the following factors does not affect the valence shell? (a) Valence principal quantum number (n) (b) Nuclear charge (Z) (c) Nuclear mass (d) Number of core electrons.

Q46- The size of isoelectronic species — F^- , Ne and Na+ is affected by

(a) nuclear charge (Z)

(b) valence principal quantum number (n)

(c) electron-electron interaction in the outer orbitals

(d) none of the factors because their size is the same.

Q47- Which one of the following statements is incorrect in relation to ionization enthalpy?

(a) Ionization enthalpy increases for each successive electron.

(b) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.

(c) End of valence electrons is marked by a big jump in ionization enthalpy.

(d) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Q48- Considering the elements B, Al, Mg, and K, the correct order of their metallic character is :

(a) B > Al > Mg > K (b) Al > Mg > B > K

(c) Mg > Al > K > B (d) K > Mg > Al > B

Q49- Considering the elements B, C, N, F, and Si, the correct order of their non-metallic character is :

(a) B > C > Si > N > F (b) Si > C > B > N > F

(c) F > N > C > B > Si m (d) F > N > C > Si > B

Q50- Considering the elements F, Cl, O and N, the correct order of their chemical reactivity in terms of oxidizing property is :

(a) F > Cl > O > N (b) F > O > Cl > N

(c) Cl > F > O > N (d) O > F > N > Cl